

Raw Socket Packet Analysis and Protocol Investigation

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1. Objective

The objective of this assignment is to understand raw socket programming, IP packet structure, protocol headers, and packet analysis using Linux and Wireshark. Students will write a C program using raw sockets, generate protocol-specific traffic, capture packets, compare results with Wireshark, and analyze packet header fields.

2. System Configuration

Operating System → **Ubuntu**

GCC Compiler → **gcc version 13.x**

Network Interface → **lo (Loopback Interface)**

3. Compile and Run Commands

Compile commands : **gcc raw_capture.c -o raw_capture**

Run commands : **sudo ./raw_capture**

4. Program Design

The raw socket program was implemented using:

```
socket(AF_INET, SOCK_RAW, IPPROTO_TCP)
```

The program captures TCP packets and extracts information from the IP and TCP headers.

IP Header Fields Captured:

- Source IP Address
- Destination IP Address
- Protocol Number
- TTL Value
- Packet Size

TCP Header Fields Captured:

- Source Port
- Destination Port
- TCP Flags

The program captures and displays 20 TCP packets.

5. Traffic Generation

Terminal 1 (Server): `nc -l 5000`

Terminal 2 (Client): `nc 127.0.0.1 5000`

Message sent: Hello Kaushal

Captured Interface: `lo` (Loopback Interface)

6. Experimental Results

Sample output from the raw socket program:

ROLL_NO=CSB24069

ASSIGNED_PROTOCOL=TCP

```

PACKET_NO=20
SRC_IP=142.251.155.119
DST_IP=172.29.115.55
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=117
PACKET_SIZE=1320
SRC_PORT=443
DST_PORT=39458
TCP_FLAGS=ACK PSH

```

Similarly, the program captured a total of 20 TCP packets.

7. Wireshark Verification

Display filter used: **ip.proto == 6**

The image shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet capture and analysis. The display filter bar at the top shows 'ip.proto == 6'. The main packet list pane displays 29 captured packets, all of which are TCP segments. The selected packet (No. 14) is highlighted in blue. The packet details pane on the right shows the structure of the selected packet: Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	50642 → 5000 [SYN, Seq=0 Win=65495 Len=0 MSS=65495 SACK PERM TSval=3089769091 TSecr=0 WS=128
2	0.000023273	127.0.0.1	127.0.0.1	TCP	74	5000 → 50642 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK PERM TSval=3089769091 TSecr=3089769091
3	0.000045859	127.0.0.1	127.0.0.1	TCP	66	50642 → 5000 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=3089769091 TSecr=3089769091
4	0.000173346	127.0.0.1	127.0.0.1	TCP	78	5000 → 50642 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=12 TSval=3089769091 TSecr=3089769091
5	0.000193984	127.0.0.1	127.0.0.1	TCP	66	50642 → 5000 [ACK] Seq=1 Ack=13 Win=65536 Len=0 TSval=3089769091 TSecr=3089769091
6	8.286809310	127.0.0.1	127.0.0.1	TCP	82	50642 → 5000 [PSH, ACK] Seq=1 Ack=13 Win=65536 Len=16 TSval=3089777378 TSecr=3089769091
7	8.286854456	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=17 Win=65536 Len=0 TSval=3089777378 TSecr=3089777378
8	23.885898764	127.0.0.1	127.0.0.1	TCP	93	50642 → 5000 [PSH, ACK] Seq=17 Ack=13 Win=65536 Len=27 TSval=3089792977 TSecr=3089777378
9	23.885950658	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=44 Win=65536 Len=0 TSval=3089792977 TSecr=3089792977
10	29.683587971	127.0.0.1	127.0.0.1	TCP	67	50642 → 5000 [PSH, ACK] Seq=44 Ack=13 Win=65536 Len=1 TSval=3089798775 TSecr=3089792977
11	29.683637290	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=45 Win=65536 Len=0 TSval=3089798775 TSecr=3089798775
12	48.045568878	127.0.0.1	127.0.0.1	TCP	108	50642 → 5000 [PSH, ACK] Seq=45 Ack=13 Win=65536 Len=42 TSval=3089817137 TSecr=3089798775
13	48.045615074	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=87 Win=65536 Len=0 TSval=3089817137 TSecr=3089817137
14	51.898360383	127.0.0.1	127.0.0.1	TCP	72	50642 → 5000 [PSH, ACK] Seq=87 Ack=13 Win=65536 Len=6 TSval=3089820990 TSecr=3089817137
15	51.898391952	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=93 Win=65536 Len=0 TSval=3089820990 TSecr=3089820990
16	60.453505291	127.0.0.1	127.0.0.1	TCP	98	50642 → 5000 [PSH, ACK] Seq=93 Ack=13 Win=65536 Len=32 TSval=3089829545 TSecr=3089820990
17	60.453544058	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=125 Win=65536 Len=0 TSval=3089829545 TSecr=3089829545
18	71.168302562	127.0.0.1	127.0.0.1	TCP	98	50642 → 5000 [PSH, ACK] Seq=125 Ack=13 Win=65536 Len=32 TSval=3089840260 TSecr=3089829545
19	71.168356562	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=157 Win=65536 Len=0 TSval=3089840260 TSecr=3089840260
20	75.849179887	127.0.0.1	127.0.0.1	TCP	85	50642 → 5000 [PSH, ACK] Seq=157 Ack=13 Win=65536 Len=19 TSval=3089844940 TSecr=3089840260
21	75.849211018	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=176 Win=65536 Len=0 TSval=3089844940 TSecr=3089844940
22	84.093042580	127.0.0.1	127.0.0.1	TCP	89	50642 → 5000 [PSH, ACK] Seq=176 Ack=13 Win=65536 Len=23 TSval=3089853184 TSecr=3089844940
23	84.093089388	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=199 Win=65536 Len=0 TSval=3089853184 TSecr=3089853184
24	96.583359934	127.0.0.1	127.0.0.1	TCP	103	50642 → 5000 [PSH, ACK] Seq=199 Ack=13 Win=65536 Len=37 TSval=3089865675 TSecr=3089853184
25	96.583401390	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=236 Win=65536 Len=0 TSval=3089865675 TSecr=3089865675
26	96.774784612	127.0.0.1	127.0.0.1	TCP	67	50642 → 5000 [PSH, ACK] Seq=236 Ack=13 Win=65536 Len=1 TSval=3089865866 TSecr=3089865675
27	96.774831850	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=237 Win=65536 Len=0 TSval=3089865866 TSecr=3089865866
28	96.920174149	127.0.0.1	127.0.0.1	TCP	67	50642 → 5000 [PSH, ACK] Seq=237 Ack=13 Win=65536 Len=1 TSval=3089866011 TSecr=3089865866
29	96.920217442	127.0.0.1	127.0.0.1	TCP	66	5000 → 50642 [ACK] Seq=13 Ack=238 Win=65536 Len=0 TSval=3089866011 TSecr=3089866011

Frame 14: 72 bytes on wire (576 bits), 72 bytes captured (576 bits) on interface lo, id 0
 Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
 Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
 Transmission Control Protocol, Src Port: 50642, Dst Port: 5000, Seq: 87, Ack: 13, Len: 6
 Data (6 bytes)

The values observed in Wireshark matched the information displayed by the raw socket program.

8. Header Analysis

1. Why is the TTL value not zero?

TTL (Time To Live) represents the maximum number of hops a packet can travel through a network. Every router decreases the TTL value by one. The TTL is not zero because the packet reached the destination before exceeding its hop limit. If TTL becomes zero, the packet is discarded and an ICMP Time Exceeded message is generated.

2. Why do packet sizes differ?

Packet sizes vary because different TCP packets contain different amounts of data. SYN and ACK packets usually contain only header information, while data packets contain additional application data, increasing the total packet size.

3. What information does the protocol field provide?

The protocol field in the IP header identifies the transport layer protocol encapsulated inside the IP packet.

Examples:

- ICMP → Protocol Number 1
- TCP → Protocol Number 6
- UDP → Protocol Number 17

In this experiment, the protocol number was 6, indicating TCP packets.

4. What could happen if the protocol field is modified?

If the protocol field is changed, the receiving device may process the packet using the wrong protocol handler or discard it completely. This can cause communication failures or incorrect interpretation of the packet.

5. What do the TCP-specific fields reveal?

The TCP source and destination ports identify the applications involved in communication.

TCP flags indicate the current state of the connection:

- SYN: Starts a connection
- SYN-ACK: Accepts a connection request
- ACK: Acknowledges received data
- PSH: Transfers application data immediately
- FIN: Terminates a connection

9. Program Enhancement

Selected Field: IP Version4

Purpose

The IP version field specifies the version of the Internet Protocol used by the packet.

```

ROLL_NO=CSB24069
ASSIGNED_PROTOCOL=TCP

PACKET_NO=1
SRC_IP=127.0.0.1
DST_IP=127.0.0.1
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=64
PACKET_SIZE=60
IP_HEADER_LENGTH=20 bytes
SRC_PORT=57212
DST_PORT=5000
TCP_FLAGS=SYN
PAYLOAD_SIZE=0 bytes

-----

PACKET_NO=2
SRC_IP=127.0.0.1
DST_IP=127.0.0.1
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=64
PACKET_SIZE=60
IP_HEADER_LENGTH=20 bytes
SRC_PORT=5000
DST_PORT=57212
TCP_FLAGS=SYN ACK
PAYLOAD_SIZE=0 bytes

-----

PACKET_NO=3
SRC_IP=127.0.0.1
DST_IP=127.0.0.1
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=64
PACKET_SIZE=52
IP_HEADER_LENGTH=20 bytes
SRC_PORT=57212
DST_PORT=5000
TCP_FLAGS=ACK
PAYLOAD_SIZE=0 bytes

-----

PACKET_NO=4
SRC_IP=127.0.0.1
DST_IP=127.0.0.1
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=64
PACKET_SIZE=66
IP_HEADER_LENGTH=20 bytes
SRC_PORT=57212
DST_PORT=5000
TCP_FLAGS=ACK PSH
PAYLOAD_SIZE=14 bytes
PAYLOAD_DATA=hello kaushal.

-----

```

Observation

The IP header length remained constant at 20 bytes, indicating a standard IPv4 header without optional fields.

10. Reflection Answers

1. Why are root privileges required for raw sockets?

Raw sockets provide direct access to network packets, including header information. Root privileges are required to prevent unauthorized users from monitoring or modifying network traffic.

2. How are raw sockets different from TCP/UDP sockets?

Raw sockets provide direct access to complete packets and allow the user to examine IP and transport layer headers. TCP/UDP sockets provide only the application data, while the operating system handles the protocol details.

3. One advantage and one limitation of raw sockets

Advantage:

Raw sockets allow detailed packet analysis and custom packet creation.

Limitation:

They require administrative privileges and are more complex to program.

4. One networking or cybersecurity application of raw sockets

Raw sockets are used in packet sniffers, network analyzers, intrusion detection systems (IDS), and security monitoring tools.

11. Conclusion

This experiment demonstrated the use of raw sockets for capturing TCP packets and analyzing IP and TCP header fields. The captured packet details were successfully verified using Wireshark, and the results from the raw socket program matched the information observed in the packet analyzer.
